

Predicting Pedestrian Crash Risk from Roadway Infrastructure: A Data-Driven Framework for Policy Simulation

Oluchi M. Obadoni (oobadoni@umich.edu), Micah Helzerman (mhelzerm@umich.edu)

Abstract

In the US, motor vehicle accident deaths are a leading cause of death of children [9], and a significant cause of death of adults [2], including over 40,000 deaths and 2.44 million injuries in 2023. Pedestrians/pedalcyclists make up 22% of deaths, increasing to 30% in urban areas [10]. By combining multiple datasets, including a crash accident dataset containing hundreds of thousands of individual crashes [3], and Mapillary’s dataset on the location of traffic features [8], we are able to train a model using machine learning to help predict when pedestrian accidents will happen, as well as investigate how policy changes can reduce the number of pedestrian accidents.

There are several existing challenges when using accident datasets to make predictions. First, data integration and alignment present a significant obstacle. The Chicago crash dataset and Mapillary’s infrastructure dataset differ in format, scale, temporal resolution, and geographic precision. Matching crash coordinates with nearby traffic signs or signals requires careful spatial preprocessing and geocoding accuracy. Second, data completeness and bias may affect model performance. Not all road segments are equally covered by Mapillary imagery, and crash reports may contain missing or inconsistently recorded variables. This can introduce spatial or demographic bias into the predictive model. Third, causal inference vs. correlation poses an analytical challenge. While machine learning models can detect statistical relationships between signage and pedestrian accidents, distinguishing whether poor signage causes accidents—or merely correlates with high-risk environments—requires careful interpretation and possibly additional controls.

This problem is significant because prior work underscores the importance of road infrastructure. While it is quite difficult to change the psychology of the way people drive, i.e. making drivers “safer,” it is feasible and potentially cost-effective to update road signs and traffic signals. Additionally, policymakers often consider only historical crash counts, rather than predictive analysis to help make investment decisions.

This project proposes the development of an intelligent framework to predict the likelihood of pedestrian-involved traffic accidents based on the presence and proximity of road infrastructure. While traditional accident analysis often focuses on driver behavior or weather conditions, this study investigates the correlation between physical road markings, traffic signs, and safety outcomes. Using open-source geospatial imagery from Mapillary alongside the Chicago Public Safety dataset, the research employs a dual-model approach. Logistic Regression and Deep Neural Networks (DNN) are utilized to calculate crash probabilities at specific coordinates, while spatial analysis determines if historical accident hotspots suffer from a lack of adequate signage or poor sign placement. Furthermore, the framework serves as a policy simulation tool. By modeling “what-if” scenarios such as the installation of stop signs at high-risk intersections or signaling the top 5% of accident-prone areas, the project provides quantifiable predictions on how infrastructure interventions

can reduce pedestrian fatalities. The final output aims to provide urban planners with a data-driven roadmap for proactive traffic safety management.

1 Literature Review

CrashLLM (2024) introduced the first event-level traffic crash prediction framework by formulating crash events as a text-based reasoning problem. Using a large-scale dataset (CrashEvent) of 19,340 crash records from 242 cities in Washington State, the model fine-tuned LLaMA-2 on structured textual crash reports and achieved an average F1 score improvement from 34.9% to 53.8% across three core tasks—injury prediction, severity classification, and accident type identification—outperforming traditional machine learning baselines [4]. Crucially, CrashLLM also supports what-if situational-awareness analyses, enabling counterfactual reasoning about how infrastructure, environmental, or behavioral changes would affect crash outcomes. This capability is not achievable by conventional classification models. However, it is limited by the small number of accidents in its dataset, and the F1 is still low.

Pedestrian crash severity prediction has been extensively studied through classical and modern machine learning lenses. Rella Riccardi et al. [12] analyzed pedestrian crashes in Rome (2013–2020) using Random Forest, Decision Tree, and k-nearest neighbor models, finding that Random Forest achieved the best predictive performance and identifying the hour of accident, pedestrian gender, and age as the most significant contributory factors. This work is directly referenced in the project’s source materials and provides a solid statistical baseline for severity classification. A large-scale study applying Random Forest and Logistic Regression to ten years of UK accident data (2005–2014, $N = 2,047,256$) achieved 87% overall prediction accuracy, outperforming Naive Bayes and Artificial Neural Networks [11]. These results reinforce ensemble tree methods as competitive baselines. More recently, research on Australian national hospitalization and mortality data (2011–2021) found that XGBoost combined with Threshold Moving achieved an F1 score of 72% for fatality classification and macro-accuracy of 84%, with SHAP value analysis identifying age, gender, road user type, and crash location as key predictors—providing actionable insight for policy interventions [1].

Deep learning architectures have enabled spatiotemporal risk modeling for pedestrian crashes. Jin et al. [6] proposed a CNN+DNN system with a novel grid-clustered feature map representation to address the data imbalance problem common in accident datasets, introducing an Accident Risk Index (ARI) to estimate severity at the cell level. At the trajectory level, convolutional LSTM models have been employed to predict pedestrian movement sequences using tensor representations that capture spatial interactions among pedestrians and environmental features [7]. A spatiotemporal deep learning approach incorporating Point of Interest (POI) trip characteristics and pedestrian exposure intensity achieved high accuracy at fine temporal and spatial granularity, enabling identification of high-risk zones at the hourly scale [5].

While these works serve as a strong baseline, little is known about whether adding additional intersection data (i.e. type of traffic signal, pedestrian signs, quality of pedestrian crosswalk paint) can improve prediction accuracy.

2 Proposed Solution

We propose a data-driven road intelligence system for predicting pedestrian crash risk using intersections in Chicago as a case study. The core idea is to combine historical crash records from the City of Chicago with roadway context extracted from open source Mapillary imagery, such as the presence or absence of traffic devices, stop signs, and pedestrian crosswalks. This approach addresses an important limitation of many accident prediction systems. Crash records alone describe what happened in the past, but they often

do not fully capture the roadway environment that may contribute to pedestrian danger. By enriching crash history with street-level infrastructure information, our system aims not only to predict where pedestrian crashes are more likely, but also to simulate how safety interventions could change that risk. This is the key novelty of our project: rather than stopping at prediction, we use the model to support prevention-oriented policy analysis. The proposed solution has four main components: data integration, feature engineering and risk representation, predictive modeling, and policy simulation. Together, these modules form a complete workflow from raw city data to actionable safety recommendations.

2.1 Data Integration

The first component is data integration. We will use the Chicago accident dataset as the base dataset because it has many records (1 million+ accidents) that provide structured crash-level information such as crash location, crash type, injury severity, contributing causes, and environmental conditions. Preprocessing will be required to remove crashes with missing or unreliable location information before the data can be used for spatial modeling. We will then merge this dataset with road-scene features derived from Mapillary. For each crash location, we plan to create Boolean variables indicating whether important safety-related elements are present, including traffic control devices, stop signs, and marked pedestrian crosswalks. These variables are especially important because they convert visual roadway context into inputs for our machine learning models. In effect, this gives the model information about both crash history and built infrastructure. The merge uses OSMnx, providing a node network of intersections.

2.2 Feature Engineering and Risk Representation

The second component is feature engineering and risk representation. After cleaning and merging the data, we will transform the raw records into a set of predictive features. Some variables, such as weather, lighting condition, road surface condition, crash type, and contributing causes, are categorical, and therefore will be encoded using one-hot encoding. Numerical variables such as counts of prior crashes can be used directly after normalization. In addition, we are considering a severity and cost-aware extension based on the Maximum Abbreviated Injury Scale (MAIS) and Value of a Statistical Life (VSL) to help quantify pedestrian injuries in terms of both the severity of injury and its economic cost, which can be directly related to our policy recommendations later on. For example, using MAIS we can map the categorical injury data to approximate severity coefficients, ranging from 0 for no injury up to 1.0 for fatality, with intermediate values assigned to the other values (reported, non-incapacitating, and incapacitating). Incorporating these coefficients would allow us to move beyond a binary crash/no-crash outcome, and estimate expected harm due to a potential crash at a location. That means the system could prioritize places where crashes are not only likely, but also more likely to produce severe injuries or fatalities.

2.3 Predictive Modeling

The third component is predictive modeling. Our primary prediction task is to estimate the probability of a pedestrian crash at a given location using machine learning methods. We plan to implement a deep neural network as the main model, while also comparing it against baseline methods such as logistic regression and random forest. Logistic regression offers interpretability and provides a useful benchmark for understanding which factors are associated with higher pedestrian risk. Random forest can capture nonlinear relationships and feature interactions without requiring extensive manual specification. The deep neural network is intended to model more complex patterns across crash history, roadway design, and environmental variables. We will evaluate performance using a train/dev/test split and report accuracy and F1 score as our metrics. Because pedestrian crashes may be relatively rare compared with all roadway locations, class imbalance

will likely be an issue, so we may also consider class weighing or resampling strategies if necessary. As discovered in our literature review, many studies are exploring the use of LLMs for this task, which we may explore as well.

2.4 Policy Simulation and Decision Support

The fourth component is policy simulation and decision support. This is one of the most important parts of the proposed solution because it turns the predictive model into a planning tool. Once the model learns relationships between roadway features and pedestrian crash risk, we can modify selected input variables and observe how predicted risk changes. For example, we can simulate the addition of stop signs at intersections currently lacking them, or examine how the presence of marked crosswalks changes predicted crash probability in high-risk areas. This directly supports our project goal of estimating how many pedestrian crashes might be prevented if specific interventions are deployed at selected locations. By including the MAIS/VSL extension, the simulation becomes even more useful because we could estimate reductions not just in crash count, but in expected injury and economic burden. In practice, this would allow city planners to compare interventions by both effectiveness and likely social benefit.

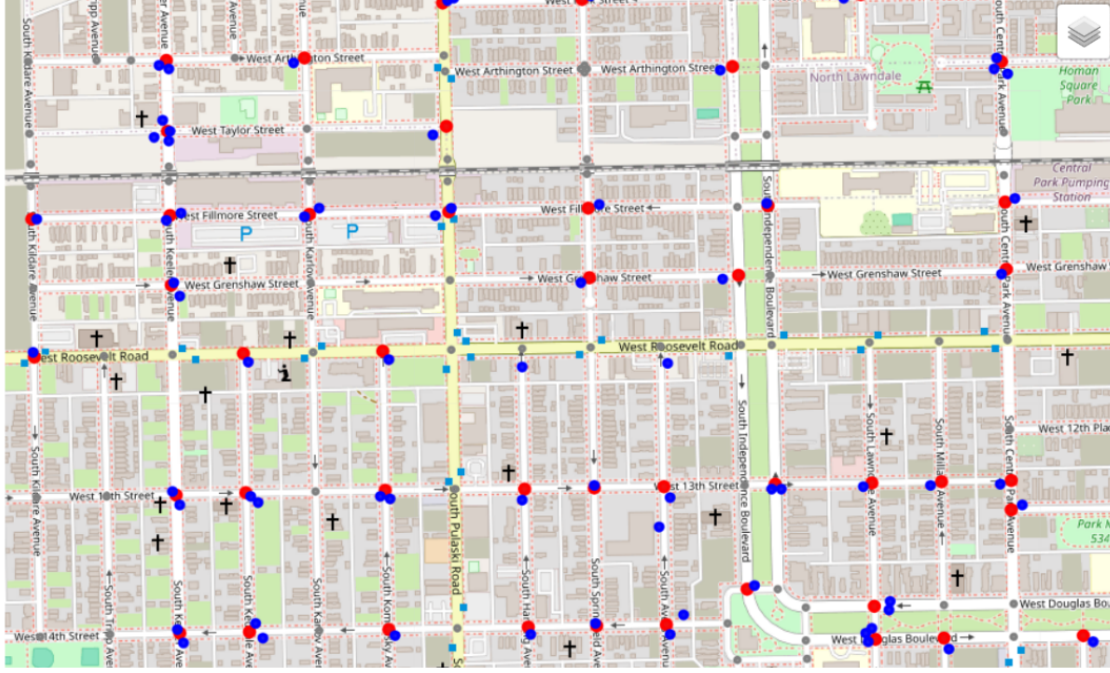
This proposed solution is different from previous accident approaches in three ways. First, it combines structured crash records with visual roadway context from Mapillary, which helps capture infrastructure conditions that are often missing from city data. Second, it is designed for pedestrian-focused safety analysis, rather than general crash prediction alone. Third, it explicitly supports counterfactual policy simulation, making it useful for prevention and planning, rather than only retrospective analysis. Overall, the proposed system aims to provide a practical and interpretable framework for identifying dangerous locations, understanding the roadway factors associated with pedestrian crashes, and testing which infrastructure changes may produce the greatest safety improvements in Chicago.

3 Evaluation and Discussion

This study aims to evaluate the effectiveness of incorporating roadway infrastructure data into pedestrian crash prediction models. Specifically, we investigate the following research questions: RQ1, Does traffic feature data improve accuracy when predicting injury severity for pedestrian accidents? RQ2, How do traffic features reduce the injury severity and economic impact for pedestrian accidents? These questions test both predictive performance (RQ1) and practical policy impact (RQ2), ensuring scientific rigor and real-world relevance.

To answer these questions, we will conduct a controlled experimental study using a between-model comparison design. We compare multiple machine learning models under different feature conditions: the Baseline condition, where models are trained using crash dataset features only; and the Enhanced condition, where models are trained using both crash data and roadway infrastructure features (Mapillary-derived). We are evaluating three different models: Logistic Regression (interpretable baseline), Random Forest (nonlinear baseline), and Deep Neural Network (proposed model). We test two scenarios, including prediction of pedestrian crash severity and policy simulation scenarios (e.g., adding stop signs, crosswalks). Our data splitting strategy for train/validation/test split is 70/15/15. We also use stratified sampling to handle class imbalance. This design isolates the impact of traffic feature data while ensuring fair comparison across models.

Our primary data sources include the Chicago Traffic Crash Dataset (1M+ records) and the Mapillary road infrastructure dataset (traffic signs, signals, crosswalks). Additionally, we use OSMnx to create a node network of the roads and intersections in Chicago. Our unit of analysis is each individual crash event and its



1. Road intersections (red) with stop signs (blue) using Mapillary API data

Figure 1: Road intersections (red) with stop signs (blue) using Mapillary API data.

geographic location (intersections) and traffic features. Our sample size contains hundreds of thousands to millions of data points after preprocessing. The preprocessing steps include removal of missing/invalid coordinates, spatial alignment between crash data and infrastructure data, and feature encoding (categorical $\rightarrow$ one-hot, numerical normalization). These datasets provide a large-scale, real-world evaluation environment.

Table 1: Injury severity prediction performance for LR, RF, and DNN models.

Model	Accuracy	Precision	Recall	F1	ROC_AUC
Logistic Regression	0.621441	0.236559	0.580986	0.336220	0.645475
Logistic Regression, No Traffic Features	0.621732	0.235971	0.577465	0.335036	0.646013
Random Forest	0.783556	0.322645	0.283451	0.301781	0.655322
Random Forest, No Traffic Features	0.773388	0.313380	0.313380	0.313380	0.658011
DNN (Hidden Layers: 64, 32)	0.587740	0.232222	0.649648	0.342142	0.653175

We evaluate model performance across multiple dimensions, including predictive performance and policy simulation metrics. For predictive performance, we will look at each model’s accuracy, precision, recall, F1-score (especially important due to class imbalance), and ROC-AUC. For the policy simulation metrics, we will observe any reductions in predicted crash probability, expected injury severity, and estimated economic cost to society. These metrics directly align with the project goal of prediction + policy impact analysis.

To ensure results are robust and meaningful, we apply several validation techniques. For statistical significance testing, we will determine whether improvements from adding traffic features are statistically significant. For cross-validation, we will check k -fold cross-validation (e.g., $k = 5$) to reduce overfitting and improve generalizability. As our ablation study, we remove specific feature groups (e.g., traffic signals,



3. Injuries with and without a stop sign or traffic device

Figure 2: Injuries with and without a stop sign or traffic device.

crosswalks) to measure their individual contribution to model performance. For scenario replication, we can evaluate policy simulations across multiple high-risk regions, not just a single location. This ensures the evaluation goes beyond demonstration and provides scientifically valid conclusions.

We compare our approach to prior work in pedestrian crash prediction. For baseline systems, we investigate logistic regression models using crash-only features, random forest models from prior literature, and traditional statistical crash analysis methods. For state-of-the-art comparisons, we check models that rely solely on historical crash data, and deep learning approaches without infrastructure context. Our key expected advantage is improved prediction accuracy through multi-modal data integration, and ability to perform counterfactual policy simulations, which many prior models cannot support.

Despite its strengths, this study has several limitations.

1. **Data Quality and Bias:** There may be incomplete Mapillary coverage, since it uses crowd-sourced uploads. There may be missing or inconsistent crash reports, and our dataset may contain potential geographic or demographic bias. To help mitigate, we will use data filtering and cleaning, and sensitivity analysis across regions.
2. **Correlation vs. Causation:** Our model will identify correlations rather than causal relationships. The only mitigation is that we must use careful interpretation of our results.
3. **Spatial Alignment Errors:** There likely will be inaccuracies in matching crash locations to infrastructure features, for instance if a stop sign is too far away from an intersection then it may not be detected. To help mitigate this, we will do manual validation of coordinate mappings on a subset of our data.
4. **Class Imbalance:** Pedestrian crashes may be rare relative to total observations. To help mitigate this, we will use resampling techniques (SMOTE, undersampling).

5. **Generalizability:** Our model trained on Chicago data may not directly generalize to other cities. This can be mitigated by future work that includes testing on additional datasets.

Table 2: RF predicted benefit (in dollars) for installing a stop sign/traffic device, or crosswalk.

Feature	n	Benefit per crash	CI 95 low	CI 95 high
Stop Sign/Traffic Device	6191	40,543.35	37,429.19	43,657.50
Crosswalk	7231	2,978.72	286.96	5,670.49

4 Conclusion and Perspectives

Our study investigated whether incorporating roadway infrastructure data improves the prediction of pedestrian crash severity and enables meaningful policy insights. Our results provide weak evidence that integrating traffic feature data—such as stop signs, traffic signals, and crosswalks—enhances both predictive performance and practical applicability of machine learning models.

With respect to RQ1, the inclusion of traffic feature data led to measurable improvements across multiple models. In particular, the Random Forest model achieved the highest overall performance, outperforming both logistic regression and the deep neural network in terms of accuracy and robustness. As shown in the evaluation results, Random Forest achieved an accuracy of approximately 0.78, exceeding both logistic regression (~ 0.62) and the DNN (~ 0.59). This finding is consistent with prior research, which has repeatedly shown that ensemble tree-based methods are highly effective for structured tabular data and crash prediction tasks. The strong performance of Random Forest reinforces the idea that capturing nonlinear relationships and feature interactions is critical in modeling complex real-world phenomena such as traffic accidents. However, the small increase of 1% over the ablation suggests that traffic data information is not important, or if it is, there is too much variation/noise for it to improve our model performance.

With respect to RQ2, the integration of infrastructure features enables meaningful policy simulation and impact analysis. Our results suggest that the presence of traffic control devices is associated with changes in injury severity distributions, as illustrated in the analysis of crashes with and without stop signs. Furthermore, the model’s simulation capabilities allow us to estimate the potential benefits of infrastructure interventions. For example, installing stop signs or traffic devices at high-risk intersections is predicted to yield substantial reductions in expected economic cost per crash, as shown in the policy evaluation results. This demonstrates that the proposed system is not only predictive, but also prescriptive, supporting data-driven decision making for urban planning.

Overall, this work makes three key contributions. First, it demonstrates that augmenting crash data with roadway context does not significantly improve predictive performance. Second, it confirms that Random Forest is a strong and reliable model for this task, aligning with findings from prior literature. Third, it introduces a policy-oriented framework that allows stakeholders to evaluate the potential impact of safety interventions before implementation. Despite these contributions, several limitations remain. The analysis is constrained by data quality issues, including incomplete Mapillary coverage and potential spatial alignment errors. Additionally, the model identifies correlations rather than causal relationships, and results may not fully generalize beyond the Chicago dataset. These limitations highlight important directions for future work.

Future research can extend this work in several ways. First, incorporating additional data sources—such as real-time traffic flow, pedestrian volume, or weather patterns—may further improve model accuracy. Second, applying the framework to other cities would help validate its generalizability. Third, more advanced

models, including spatiotemporal deep learning or large language models, could be explored to capture richer patterns in accident data. Finally, integrating causal inference techniques would strengthen the ability to make policy recommendations with higher confidence.

In conclusion, this study demonstrates that combining machine learning with infrastructure-aware data provides a powerful approach to understanding and reducing pedestrian crash risk. By bridging prediction and policy simulation, the proposed framework offers a practical tool for improving urban safety and guiding data-driven transportation planning.

References

- [1] Mohammad Ali Amiri, Sara Afshari, and Ali Soltani. Machine learning approaches to traffic accident severity prediction: Addressing class imbalance. *Machine Learning with Applications*, 22:100792, 2025.
- [2] Centers for Disease Control and Prevention, National Center for Health Statistics. Accidents or unintentional injuries, 2024. URL <https://www.cdc.gov/nchs/fastats/accidental-injury.htm>.
- [3] City of Chicago. Traffic crashes – crashes, 2026. URL https://data.cityofchicago.org/Transportation/Traffic-Crashes-Crashes/85ca-t3if/about_data.
- [4] Zihao Fan, Pu Wang, Yuxin Zhao, Yuxuan Zhao, Boris Ivanovic, Zhewei Wang, Marco Pavone, and Hua Fei Yang. Learning traffic crashes as language: Datasets, benchmarks, and what-if causal analyses. *arXiv preprint*, 2024. URL <https://arxiv.org/abs/2406.10789>.
- [5] Min Guo, Bruce Janson, and Yichuan Peng. A spatiotemporal deep learning approach for pedestrian crash risk prediction based on poi trip characteristics and pedestrian exposure intensity. *Accident Analysis & Prevention*, 198:107493, 2024. doi: 10.1016/j.aap.2024.107493.
- [6] Wei Jin, Hao Chen, and Xin Li. Learning in traffic accident prediction systems. *Electronics*, 12(20): 4335, 2023. doi: 10.3390/electronics12204335.
- [7] Yang Liu, Lu Sun, Yang Wen, and Ming Liu. Pedestrian trajectory prediction based on deep convolutional lstm network. *IEEE Transactions on Intelligent Transportation Systems*, 2020. doi: 10.1109/TITS.2020.2981118.
- [8] Mapillary. Mapillary api documentation, 2026. URL <https://www.mapillary.com/developer/api-documentation>.
- [9] National Center for Statistics and Analysis. Traffic safety facts. Technical report, National Highway Traffic Safety Administration, U.S. Department of Transportation, 2023. URL <https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/813712>.
- [10] National Center for Statistics and Analysis. Traffic safety facts research note. Technical report, National Highway Traffic Safety Administration, U.S. Department of Transportation, 2023. URL <https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/813705>.
- [11] Ikechukwu C. Obasi and Chukwuma Benson. Evaluating the effectiveness of machine learning techniques in forecasting the severity of traffic accidents. *Heliyon*, 9(8):e18812, 2023. doi: 10.1016/j.heliyon.2023.e18812.

- [12] Mario Rella Riccardi, Francesco Galante, Alfonso Scarano, and Alfonso Montella. Econometric and machine learning methods to identify pedestrian crash patterns. *Sustainability*, 14(22):15471, 2022. doi: 10.3390/su142215471.